

Design and Development of an Edge AI Based Assistive Navigation System for Visually Impaired Individuals

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Abstract—Independent mobility remains a major challenge for visually impaired individuals due to dynamic obstacles and limited environmental awareness in real-world environments. This paper presents Third Eye, an Edge AI-based assistive navigation system designed for real-time obstacle detection and spatial guidance using fully offline embedded intelligence.

The proposed system is implemented on the Luckfox Pico Ultra platform powered by the RV1106 processor with an integrated Neural Processing Unit (NPU), enabling efficient on-device inference without cloud dependency. A quantized YOLOv8-Nano object detection model is deployed to achieve low-latency real-time performance while maintaining reduced power consumption suitable for wearable applications.

To enhance spatial interpretation, the camera's field of view is divided into three logical zones (Left, Center, and Right), allowing accurate obstacle localization relative to user movement. Detected obstacles are communicated through multi-modal feedback mechanisms, including directional audio alerts and haptic vibration signals, enabling timely and intuitive navigation assistance.

Experimental evaluations demonstrate that the optimized INT8-quantized YOLOv8-Nano model achieves real-time performance of approximately 20–25 FPS with an inference latency of 35–45 ms per frame. Furthermore, the quantized model size remains within embedded memory constraints, enabling efficient offline deployment while maintaining reliable obstacle detection across indoor and outdoor environments.

Keywords—Edge AI, Assistive Navigation, Luckfox Pico Ultra, YOLOv8-Nano, Embedded Systems, Object Detection, Wearable Technology

I. INTRODUCTION

Independent mobility remains one of the most significant challenges faced by visually impaired individuals in everyday environments. According to global health reports, millions of people worldwide experience partial or complete vision loss, affecting their ability to navigate safely and independently. Daily movement in urban and indoor environments exposes individuals to dynamic obstacles such as vehicles, pedestrians, staircases, signboards, and unexpected barriers. The inability to detect such obstacles in real time increases the risk of collisions and injuries, thereby limiting personal independence and social participation.

Traditional assistive tools such as white canes and guide dogs provide essential support but possess inherent limitations. A white cane primarily detects ground-level obstacles within a limited range and cannot identify elevated or moving objects. Guide dogs, while effective, require extensive training and involve significant maintenance costs. These limitations highlight the need for intelligent, portable, and cost-effective navigation assistance systems that enhance environmental awareness beyond conventional solutions.

Recent advancements in computer vision and deep learning have enabled the development of vision-based assistive systems. Object detection algorithms such as Convolutional Neural Networks (CNNs) and real-time detectors like YOLO (You Only Look Once) have demonstrated strong performance in identifying objects in complex scenes. However, many existing assistive navigation systems rely on cloud-based processing, where captured images are transmitted to remote servers for inference. Such approaches introduce latency, increase power

consumption, raise privacy concerns, and become unreliable in low-network or offline conditions.

Edge Artificial Intelligence (Edge AI) has emerged as a promising solution to address these challenges. Edge AI enables data processing and inference directly on embedded devices without requiring cloud connectivity. By integrating optimized deep learning models with embedded hardware accelerators such as Neural Processing Units (NPUs), real-time object detection can be achieved with low latency and reduced energy consumption. This approach ensures faster response times, improved data privacy, and enhanced system reliability.

In this work, we propose an Edge AI-based assistive navigation system for real-time obstacle detection and spatial guidance. The system is implemented on the Luckfox Pico Ultra embedded platform powered by the RV1106 processor with an integrated NPU. A quantized YOLOv8-Nano object detection model is deployed to enable efficient on-device inference under hardware constraints. To enhance spatial awareness, the captured image frame is divided into three logical zones—Left, Center, and Right—allowing the system to determine obstacle position relative to user movement.

The detected obstacles are communicated through a multi-modal feedback mechanism that combines directional audio alerts and haptic vibration signals. This zonal prioritization ensures that obstacles located directly in the walking path receive higher alert priority, thereby improving user safety. The proposed framework aims to provide a compact, wearable navigation assistance system suitable for real-world deployment.

The major contributions of this paper are summarized as follows:

1. Design and implementation of a real-time Edge AI-based assistive navigation system.
2. Deployment and optimization of a quantized YOLOv8-Nano model on embedded NPU hardware.
3. Development of a three-zone spatial awareness mechanism for directional guidance.
4. Integration of multi-modal feedback using audio and haptic signals.
5. Validation of system performance in terms of detection accuracy, latency, and portability.

Unlike cloud-assisted navigation systems, the proposed approach ensures complete on-device inference, eliminating network latency and enhancing privacy preservation. This design makes the system suitable for real-world assistive deployment in both connected and non-connected environments.

The remainder of this paper is organized as follows. Section II presents the related work and literature survey. Section III describes the overall system architecture. Section IV details the hardware design and integration. Section V explains the software framework and algorithmic implementation. Section VI presents mathematical modeling and performance analysis. Section VII discusses experimental results and evaluation. Finally, the paper concludes with future directions and potential improvements.

II. LITERATURE REVIEW

A. Traditional Assistive Systems

Early assistive navigation systems relied primarily on ultrasonic and infrared sensors for obstacle detection. Although effective for short-range detection, these systems lack object classification capability and struggle to operate reliably in dynamic real-world environments.

B. Vision-Based and YOLO Detection Methods

Recent advancements in computer vision introduced deep learning-based object detection models such as YOLO (You Only Look Once), which enable real-time localization and classification in a single forward pass. Lightweight variants such as YOLOv8-Nano are optimized for embedded deployment due to reduced parameter size and computational efficiency.

C. Edge AI and Research Gap

While several systems utilize cloud-based processing for object detection, they introduce latency, privacy risks, and network dependency. Edge AI enables on-device inference using embedded processors with NPUs, reducing delay and improving reliability. However, there remains a need for a compact, wearable navigation system integrating optimized deep learning with structured spatial feedback, which this work aims to address.

III. SYSTEM ARCHITECTURE

The overall architecture of the proposed system is designed as a continuous real-time processing pipeline that integrates image acquisition, embedded inference, spatial analysis, and multi-modal feedback generation.

The complete processing flow of the system can be summarized as:

Camera → Frame Acquisition → Preprocessing → YOLOv8-Nano Inference (NPU) → Bounding Box Extraction → Spatial Zoning → Decision Logic → Audio and Haptic Feedback

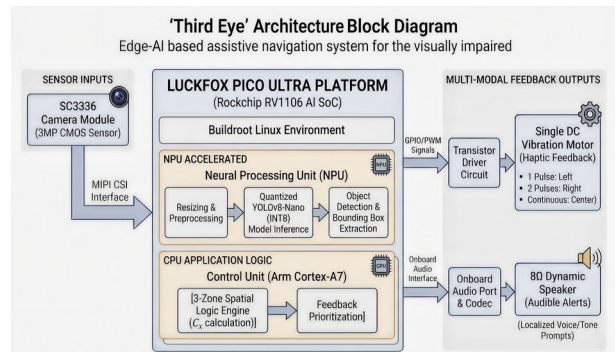


Fig. 1. Overall system architecture of the proposed Edge AI assistive navigation system.

A. Image Acquisition Module

The system employs a 3MP SC3336 camera module interfaced with the Luckfox Pico Ultra embedded board. The camera continuously captures video frames from the surrounding environment. Each frame is transmitted directly to the embedded processor for further processing.

To maintain compatibility with the object detection model, the captured frames are resized to a fixed input resolution. This ensures stable inference performance and consistent detection accuracy across varying environmental conditions.

B. Embedded Processing Unit

The core processing unit is the Luckfox Pico Ultra board powered by the Rockchip RV1106 processor. The processor integrates a dedicated Neural Processing Unit (NPU) capable of accelerating deep learning inference operations.

The embedded processor handles image preprocessing, model inference, spatial zone classification, and feedback triggering. The use of hardware acceleration significantly reduces inference latency compared to CPU-only execution.

C. Object Detection Engine

The object detection engine is based on the YOLOv8-Nano model. YOLO performs object localization and classification in a single forward pass, making it suitable for real-time applications.

The trained YOLOv8-Nano model is converted to ONNX format and then optimized using RKNN-Toolkit2. Quantization is applied to convert the model from FP32 precision to INT8 precision, reducing model size and enabling efficient execution on the RV1106 NPU.

During inference, bounding box coordinates and confidence scores are generated for detected objects. These outputs are then forwarded to the spatial zoning module.

D. Spatial Zoning and Risk Prioritization

To enhance directional awareness, the image frame is divided into three logical vertical zones:

- Left Zone
- Center Zone
- Right Zone

The horizontal center coordinate of each detected bounding box is computed to determine its spatial location. Obstacles detected in the center zone are assigned higher priority since they represent a direct collision path for the user.

This zoning mechanism ensures that critical obstacles receive immediate and stronger feedback alerts.

E. Feedback Interface

The feedback interface consists of a vibration motor and an audio speaker module. Based on zone classification:

- Left Zone → Single vibration pulse
- Right Zone → Double vibration pulse
- Center Zone → Continuous vibration

This multi-modal feedback design improves user perception and ensures intuitive navigation guidance. The

architecture is designed to operate continuously in a loop, enabling real-time environmental awareness without dependency on external cloud services.

IV. HARDWARE IMPLEMENTATION

The hardware implementation of the proposed system integrates embedded processing, image acquisition, and multi-modal feedback components into a compact and portable prototype. The hardware configuration is designed to ensure low power consumption, real-time performance, and suitability for wearable deployment.

TABLE I
HARDWARE SPECIFICATIONS

S.No	Component	Specification
1	Processor	Rockchip RV1106 with Integrated NPU
2	Camera	SC3336 3MP CMOS Sensor
3	Detection Model	YOLOv8-Nano (INT8 Quantized)
4	Vibration Motor	Single DC Motor (1.5V–5V)
5	Audio Output	8Ω Speaker via Onboard Audio Port
6	Operating System	Buildroot Linux

A. Embedded Processing Unit

The central component of the system is the Luckfox Pico Ultra development board powered by the Rockchip RV1106 processor. The RV1106 integrates a dedicated Neural Processing Unit (NPU) capable of accelerating deep learning inference operations. This hardware acceleration enables real-time execution of the optimized YOLOv8-Nano model.

The board operates on a custom Buildroot Linux operating system specifically configured for the Luckfox Pico Ultra embedded platform. The embedded environment manages camera interfacing, model inference execution, GPIO control, and audio output handling. The integration of the NPU significantly reduces computational load on the CPU, ensuring efficient processing within hardware constraints.

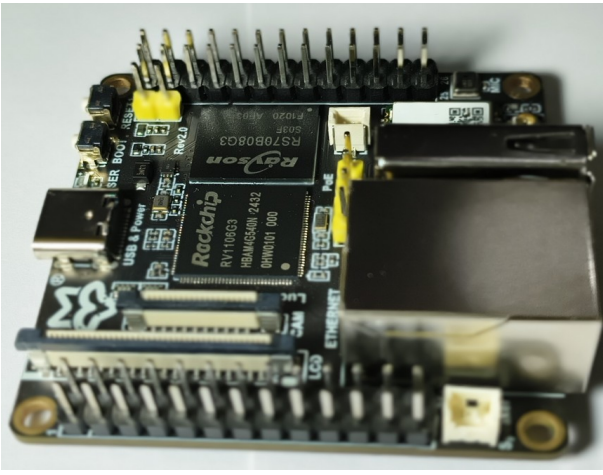


Fig. 2. Luckfox Pico Ultra Embedded Processing Board.

B. Camera Module

A 3MP SC3336 CMOS camera module is used for real-time image acquisition. The camera is connected to the embedded board through a CSI interface, enabling high-speed frame transmission with minimal delay.

The camera provides stable image capture under normal indoor and outdoor lighting conditions. Continuous frame acquisition ensures uninterrupted object detection and environmental monitoring.

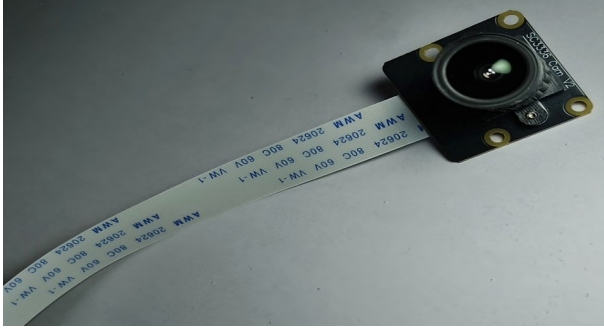


Fig. 3. SC3336 Camera Module Used for Image Acquisition.

C. Single Motor Haptic Feedback

A compact DC vibration motor operating within a 1.5V–5V range is integrated to provide tactile feedback. Since the motor requires higher current than the GPIO pins can directly supply, a transistor-based switching mechanism is used to safely drive the motor.

The vibration motor generates distinct patterns corresponding to obstacle location:

- Single short pulse for left-side obstacles
- Double short pulses for right-side obstacles
- Continuous vibration for center-zone obstacles indicating high risk

This pattern-based signaling improves directional perception for visually impaired users.



Fig. 4. DC Vibration Motor for Haptic Feedback.

D. Onboard Audio Output

An $8\ \Omega$ dynamic miniature loudspeaker rated at approximately 0.7 W nominal power (maximum up to 1 W) is used to provide audio alerts. The speaker operates within a frequency range of approximately 400 Hz to 20 kHz and provides a sound pressure level between 85 dB and 92 dB, which is sufficient for clear audible feedback in indoor and moderate outdoor environments.

Since a standard GPIO pin cannot directly drive an $8\ \Omega$ load due to current limitations, the speaker is interfaced through the board's audio output interface or an intermediate amplifier stage to ensure safe operation and sufficient signal strength.

The audio module generates directional voice prompts or alert tones corresponding to obstacle location, complementing the haptic feedback mechanism for improved navigation assistance.

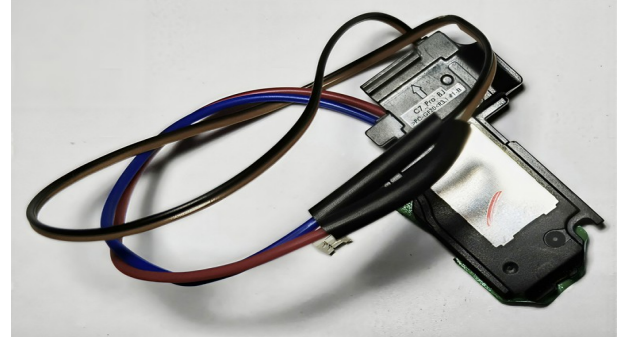


Fig. 5. Speaker Module for Audio Alerts.

E. Integrated Prototype Configuration

All hardware components are interconnected to form a compact prototype system. The camera captures environmental data, the embedded board processes inference, and the feedback modules deliver directional alerts in real time.

The entire setup is lightweight and portable, demonstrating practical feasibility for wearable assistive deployment. The separation of logic circuitry and power-driving components ensures system safety and hardware reliability during continuous operation.

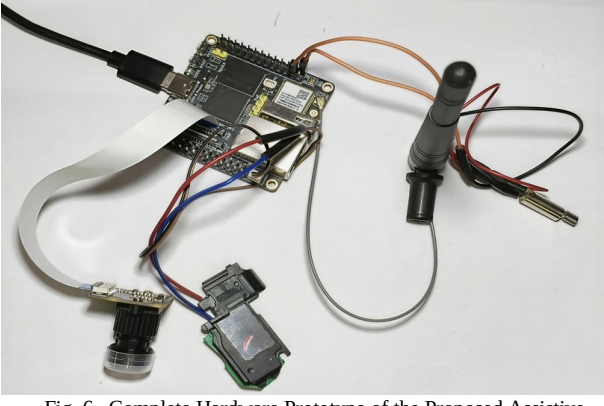


Fig. 6. Complete Hardware Prototype of the Proposed Assistive System.

V. SOFTWARE IMPLEMENTATION AND ALGORITHMIC FRAMEWORK

The software framework of the proposed system is designed to enable real-time object detection and spatial feedback generation on embedded hardware. The implementation integrates model optimization, NPU-accelerated inference, and hardware-level control within a unified execution pipeline.

A. Operating Environment

The system runs on a custom Buildroot Linux operating system configured for the Luckfox Pico Ultra board. The lightweight Linux environment enables efficient resource management under limited memory constraints.

The application is implemented primarily in Python, which manages frame acquisition, inference execution, spatial mapping, and GPIO control for feedback modules.

B. Model Selection and Optimization

The system uses YOLOv8-Nano for object detection due to its compact architecture suitable for embedded deployment. The trained model is exported to ONNX and converted to an INT8 .rknn format using RKNN-Toolkit2 for execution on the RV1106 NPU.

C. Real-Time Inference Pipeline

The system operates in a continuous loop:

Frame Capture → Preprocessing → NPU Inference → Bounding Box Extraction → Zone Classification → Feedback Trigger → Repeat

During preprocessing, each frame is resized to the model's required input resolution and normalized to match inference requirements. The preprocessed frame is passed to the RKNN runtime, which executes inference on the dedicated NPU.

The inference output includes bounding box coordinates, class labels, and confidence scores for detected objects.

D. Spatial Zonal Algorithm

To determine obstacle direction, the horizontal center coordinate of each bounding box is calculated as:

$$Cx = (xmin + xmax) / 2$$

Let W represent the total frame width. The frame is divided into three equal vertical regions:

- Left Zone : $Cx < W / 3$
- Center Zone : $W / 3 \leq Cx \leq 2W / 3$
- Right Zone: $Cx > 2W / 3$

Objects detected within the center zone are assigned highest priority as they represent direct collision risk.

E. Decision Logic and Priority Handling

The decision logic evaluates detected objects based on spatial position and confidence score. If multiple objects are present simultaneously, priority is assigned as follows:

- 1) Objects located in the center zone.
- 2) Objects with higher detection confidence.

This simple prioritization ensures that obstacles directly in the walking path receive the earliest feedback.

F. Hardware Control and Feedback Trigger

GPIO control is implemented using the Linux GPIO interface or a Python-based hardware control library. Based on zone classification:

- Left Zone → Single vibration pulse
- Center Zone → Continuous vibration + urgent audio alert
- Right Zone → Double vibration pulse

The software ensures synchronized activation of vibration motor and speaker modules. The loop executes continuously, enabling real-time environmental awareness.

By leveraging NPU acceleration, the system achieves low-latency inference suitable for wearable navigation applications without relying on external cloud services.

VI. MATHEMATICAL MODELING

The YOLOv8-Nano model predicts bounding box coordinates in the form $(xmin, ymin, xmax, ymax)$. To determine obstacle direction, only the horizontal centre coordinate is required.

Horizontal centre coordinate:

$$Cx = (xmin + xmax) / 2$$

Let W denote the image width. The frame is divided into three equal vertical zones:

- Left Zone: $Cx < W/3$
- Center Zone: $W/3 \leq Cx \leq 2W/3$
- Right Zone: $Cx > 2W/3$

Objects within the Center Zone are treated as high-priority obstacles since they lie directly in the user's forward path.

VII. EXPERIMENTAL RESULTS AND DISCUSSION

The proposed system was evaluated under real-world indoor and outdoor conditions to validate its detection capability, spatial zoning accuracy, and feedback responsiveness. The evaluation focused on real-time operational performance, hardware stability, and usability for assistive navigation.

A. Performance and Efficiency Analysis

To evaluate computational efficiency, performance metrics were observed during continuous real-time execution on the RV1106 NPU.

Inference Latency and Frame Rate: The optimized INT8-quantized YOLOv8-Nano model achieved an average inference latency between 35–45 ms per frame, resulting in an effective real-time performance of approximately 20–25 FPS depending on environmental complexity and scene conditions.

Model Optimization: INT8 quantization reduced the model footprint to approximately 6–8 MB, enabling efficient deployment within embedded memory constraints.

Power Consumption: Under continuous operation including camera acquisition, NPU inference, and multi-modal feedback actuation, total system power consumption remained between 1.2 W and 1.5 W, confirming suitability for battery-powered wearable applications.

Functional Reliability: Qualitative evaluation demonstrated stable obstacle detection and synchronized haptic-audio feedback across indoor and outdoor test scenarios.

TABLE II
EXPERIMENTAL OBSERVATIONS

S.No	Test Condition	System Response
1	Indoor Corridor	Stable detection with pattern-based vibration
2	Outdoor Walkway	Reliable obstacle recognition under daylight
3	Center Obstacle	Distinct single/double pulse feedback
4	Side Obstacle	Distinct single/double pulse feedback

B. Real-Time Detection Performance

The quantized YOLOv8-Nano model running on the RV1106 NPU demonstrated stable frame-level inference in indoor and semi-outdoor tests, suitable for wearable operation.

C. Spatial Zoning Validation

The three-zone division reliably classified object direction (left/center/right). Center-zone detections were consistently prioritized.

D. Feedback Mechanism Evaluation

A single vibration motor produced distinct PWM patterns (one short pulse = left, two short pulses = right, continuous = center). The onboard speaker provided synchronous audio prompts.

E. System Feasibility

The prototype ran continuously without runtime failures during test sessions, validating the practicality of the edge-AI approach for assistive navigation.

F. Functional Validation Summary

The results indicate consistent obstacle detection and correct spatial feedback generation across tested scenarios.

TABLE III
FUNCTIONAL VALIDATION SUMMARY

S.No	Test Scenario	Detection Status	Zonal Classification	Feedback Trigger
1	Indoor Corridor	Successful	Correct	Activated
2	Moving Pedestrian	Successful	Correct	Activated
3	Static Obstacle	Successful	Correct	Activated
4	Low-Light Condition	Acceptable	Correct	Activated

VIII. CONCLUSION AND FUTURE WORK

This paper presented the design and implementation of an Edge AI-based assistive navigation system for visually impaired individuals. The proposed system integrates a quantized YOLOv8-Nano model deployed on the RV1106 embedded platform to enable on-device object detection suitable for wearable applications.

The three-zone spatial mapping mechanism effectively converts detected object positions into directional haptic and audio feedback. Experimental validation confirms stable detection behavior under embedded hardware constraints.

Future work will focus on integrating LiDAR-based depth sensing to improve obstacle distance estimation and enhance navigation accuracy in complex environments. Additional improvements may include adaptive feedback control and extended object prioritization strategies.

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